

COMPOUND PP

Superior Weatherability and Heat-Resistant
PP Compound Resins

- ▶ TB52U
- ▶ TB71HW
- ▶ TB72P
- ▶ TB72W
- ▶ TB81W

● Description

Super weatherability and heat-resistant compound PP are ideally suited for use in air-conditioner exterior parts and other outdoor purposes.

We have significantly improved strength and heat-resistance by adding talc, the inorganic filler, to polypropylene. These products are made with a variety of base PP, such as HIPP (High Isotactic Polypropylene), providing our customers with the best quality products of their kind on the market.

In addition to the characteristics noted above, Hanwha Total's super weatherability, heat-resistant compound PP exhibit excellent impact-resistance, long-term heat-resistance, resistance to thermo-discoloration and anti-static property, and dimensional stability.

● Characteristics

- ▶ Excellent weatherability:

TB52U has received certification for environmental exposure, meeting the UL746C F1 and UL484 standards,

- ▶ High heat-resistance and dimensional stability;.
- ▶ Superior rigidity and impact-resistance;
- ▶ Superior processability.

● Applications

- ▶ Outdoor electric and electronic parts such as air-conditioner exterior parts (housing, side covers, grills, top plates , etc.).
- ▶ Outdoor industrial use such as transport containers.

● Physical Properties

▶ Resin Properties

Properties	Test Method	Condition	Unit	TB52U	TB71HW	TB72W	TB72P	TB81W
Melt Index	ASTM D1238	230 °C	g/10min	12.0	20.0	24.0	23.0	40.0
Density	ASTM D1505	-	g/cm ³	1.04	1.00	1.05	1.05	0.94
Tensile Strength	ASTM D638	50mm/min	kg/cm ²	350	280	310	270	310
Elongation at Break			%	60	30	50	50	50
Flexural Stiffness	ASTM D747	5mm/min	kg/cm ²	500	430	440	420	460
Flexural Modulus	ASTM D790			27,000	24,000	24,000	22,000	23,000
Izod Impact Strength	ASTM D256	23 °C	kg.cm/cm	8.5	7.0	5.0	6.5	5.0
Rockwell Hardness	ASTM D785	Rockwell	R-Scale	93	100	90	85	89
Heat Deflection Temp.	ASTM D648	4.6kg/cm ²	°C	139	135	133	125	115
Shrinkage	HANWHA TOTAL	2mmt	%	1.0~1.4	1.2~1.6	1.0~1.4	1.0~1.4	1.3~1.7

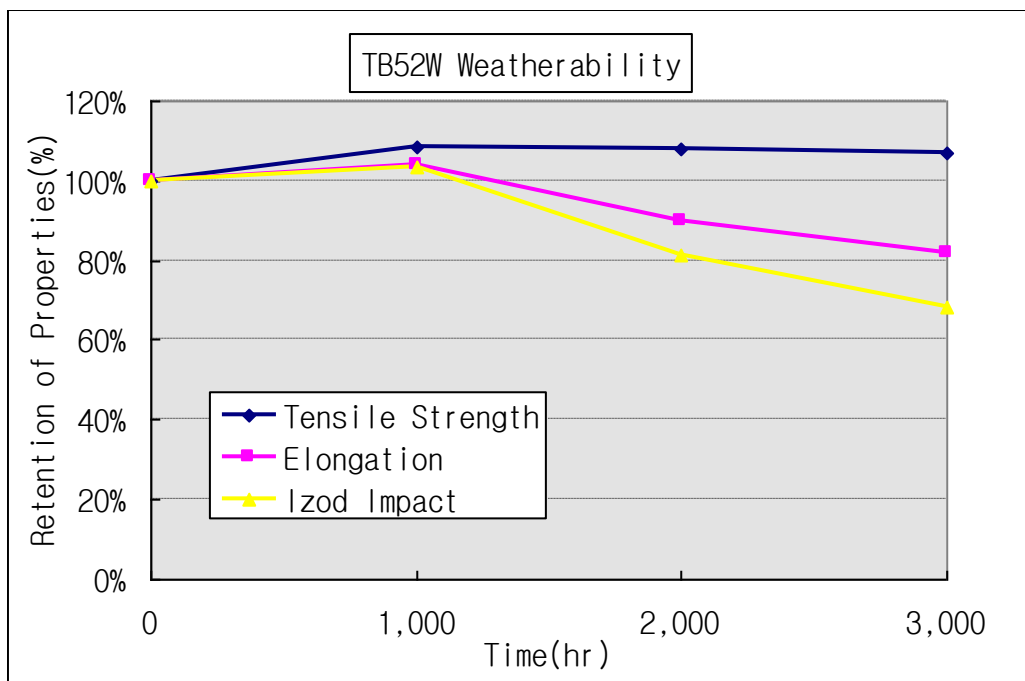
* Data shown above are representative values for reference purposes only, and not to be construed as specifications.

● Weatherability

The following graphs show the result of tests conducted using a weather-o-meter. Hanwha Total's super weatherability, heat-resistant compound PP exhibits top-of-the-line quality, and TB52U is certified UL746C F1 and UL484 for environmental-exposure

- ▶ Test conditions: Xenon-arc type; Black panel temperature of 63 °C, 0~3,000hr aging
- ▶ Test Grade: TB52U
- ▶ Test Results:

Properties	Method	Unit	Aging time (hr)			
			0	1,000	2,000	3,000
Tensile strength	ISO 527	kg/cm ²	265	287	286	283
Elongation	ISO 178	%	50	52	45	41
Izod Impact strength	ISO 180	kg.cm/cm	5.4	5.58	4.4	3.7
Δ E			-	0.55	0.72	1.02



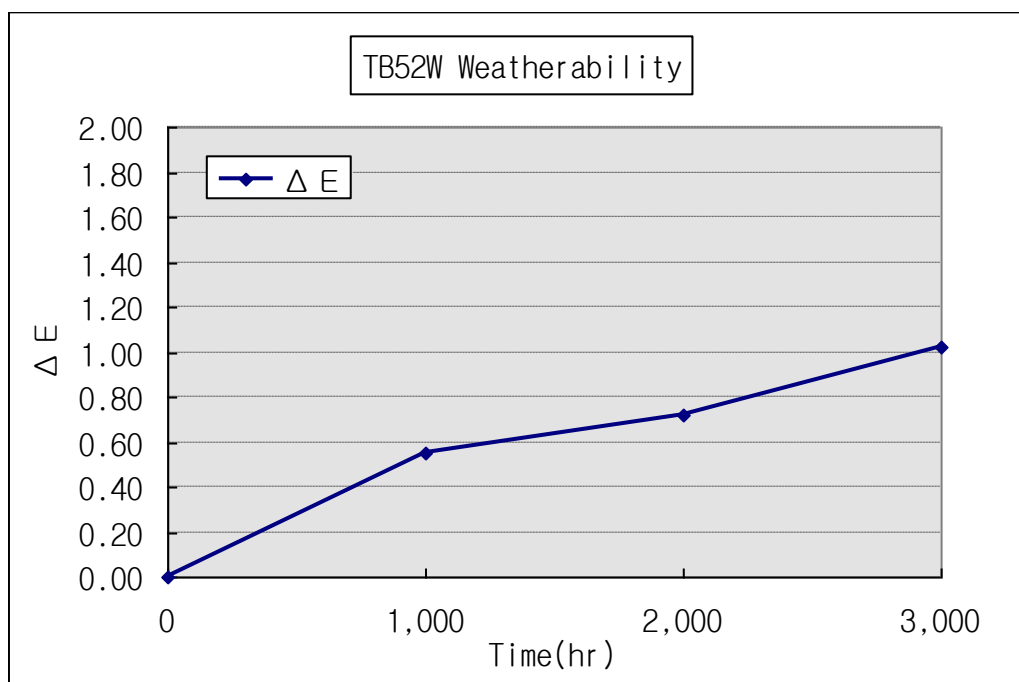


Figure. Weatherability of TB52W, Samsung's super weatherability and heat-resistant compound PP

● **General Processing Guide**

▶ Typical Processing Conditions are as follows.

- Feeding zone temperature: 180 ~ 200 °C
- Plasticizing zone temperature: 190 ~ 210 °C
- Measuring zone temperature: 200 ~ 220 °C
- Mold temperature: 40 ~ 80 °C
- Screw rpm: 30 ~ 80
- Injection pressure: 600 ~ 1,000 kg/cm²
- Holding pressure: 400 ~ 900 kg/cm²

Actual injection conditions must be adjusted according to the types of injection machines and sizes of molds and parts.

● **Food Contact Application**

- ▶ Hanwha Total TB52U, TB71HW, TB72P, TB72W, TB81W are not intended for use in food-contact and medical applications either.
- ▶ In case you might need additional technical or regulatory information, please contact Hanwha Total Composite Development Team.

● **Other Information**

The information in this document can be used for reference only, not to be construed as specification. Customers are responsible for determine whether our product and information is suitable for their particular purpose and for the compliance with related law.

Hanwha Total assumes no obligation or liability for the information in this document.